Post-vaccination rebounds in the SIR model

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1 Theoretical honeymoon period in the SEIR model

We will consider how reductions in vaccination rates can lead to rebounds in disease prevalence in a SEIR (Susceptible, Exposed, Infected, Recovered) model. In particular, we will focus on the post-vaccination "honeymoon", or how long it takes for the disease to rebound after vaccination rates start declining.

To begin, we will consider the simplest case where (a) a population has been under the vaccination regime for sufficiently long period of time to be at equilibrium, and (b) that the pathogen is locally extinct. Let v_0 be the vaccination rate during the period of herd immunity, and let v be the vaccination rate after the decline begins.

In the SEIR model, we know there are two possible equilibria:

1. disease-free equilibrium, where $S^* = v_0$
2. endemic equilibrium, where $S^* = \frac{1}{\beta}(\mu + \gamma + \frac{mu(\mu+\gamma)}{\sigma})$

Here, we are only considering the first case where a population is above the herd immunity threshold, $1/R_0$, because otherwise there is no honeymoon period. We are interested in the duration of the honeymoon period, which we hypothesize will depend on a number of factors including the R_0 of the pathogen, the equilibrium number of susceptibles S^*, the birth rate μ, the new vaccination coverage v under decline, and the timing of importations. Because we cannot control the timing of importations, we ignore this piece and define the honeymoon period as the time it takes for $R_e = 1/R_0 * S > 1$. In other words, the time it takes for R_e to cross 1 is a lower bound on the honeymoon period, where any importation after this point will start an outbreak.

We can calculate analytically the time it takes for R_e to cross 1, under assumptions of local extinction (i.e., $I = 0$) and a constant vaccination rate, v. In this case, we have a differential equation for the number of susceptibles, S, which we can solve using the following steps:

$$\frac{dS}{dt} = \mu(1-v) - \mu S \tag{1}$$

$$e^{\mu t}\Big(\frac{dS}{dt} + \mu S\Big) = e^{\mu t}\mu(1-v) \tag{2}$$

$$\int e^{\mu t}\Big(\frac{dS}{dt} + \mu S\Big)dt = \int e^{\mu t}\mu(1-v)dt \tag{3}$$

$$\int e^{\mu t}\frac{dS}{dt} + e^{\mu t}\mu S dt = \int e^{\mu t}\mu(1-v)dt \tag{4}$$

$$e^{\mu t}S = (1-v)e^{\mu t} + C \tag{5}$$

$$S(t) = (1-v) + Ce^{-\mu t} \tag{6}$$

and solving $S(0)$ gives us $C = S_0 - (1-v)$. Thus, we have:

$$S(t) = (1-v) + (S_0 - (1-v))e^{-\mu t}$$

We can find the time t it takes for $R_e = 1/R_0 * S(t) > 1$, which yields

$$t = \frac{-1}{\mu}\log\Big(\frac{1/R_0 - (1-v)}{S_0 - (1-v)}\Big)$$

We can also rewrite this equation to show us the relationship between prior vaccinoation coverage and new vaccination coverage, that will yield equivalent honeymoon periods. When vaccination rates are above the herd immunity threshold, $S_0 = v_0$ the pre-drop vaccination coverage, so

$$t = \frac{-1}{\mu}\log\Big(\frac{1/R_0 - (1-v)}{v_0 - (1-v)}\Big) \tag{7}$$

$$v_0 = v - e^{-\mu t}\Big(1/R_0 - (1-v)\Big) \tag{8}$$

$$v_0 = e^{-\mu t}(1 - 1/R_0) + (1 + e^{-\mu t})v \tag{9}$$

2 Unity beta and the effect of age structure

2.1 Unity beta calculation

Age structured dynamics are known to be important for a range of infectious disease, and we will consider how this affects the honeymoon period. In particular, we will use a theoretical quantity called "unity beta" to explore these effects.

First we outline the unity beta calculation. Consider an age-structured transmission process that is governed by heterogeneous contacts between age groups, where the number of new infections in the age structured model is

$$\text{age-structured infections} = \frac{\beta}{N}\sum_{i=1}^{n}\sum_{i=j}^{n}\theta_{i,j}S_iI_j \tag{10}$$

$$= \frac{\beta}{kN}\sum_{i=1}^{n}\sum_{i=j}^{n}\theta_{i,j}S_iI_j \tag{11}$$

where $\theta_{i,j}$ is the contact rate between age groups i and j and k is the quantity called "unity beta" (i.e., the scalar on β such that the number of new infections in the age-structured model is equal to the number of new infections in a non-age-structured model). Then,

$$k = \frac{\sum_{i=1}^{n} \sum_{i=j}^{n} \theta_{i,j} S_i I_j}{\sum_{i=1}^{n} S_i \sum_{i=j}^{n} I_j}$$

where $1/k$ is the multiplier on the transmission rate. Thus, if $1/k > 1$ then $\lambda_{\text{hom}} > \lambda_{\text{het}}$ and the age-structured dynamics are slow relative to the homogenous dynamics.

However, in the analysis, we do something slightly different. For the same transmission rate, different WAIFW matrices can yield different numbers of infections at equilibrium. Thus, we calculated a scalar c such that the number of infections at equilibrium in the age-structured model is equal to the number of infections in the non-age-structured model (i.e., $\beta_{\text{hom}} = c\beta_{\text{het}}$).

If we calculate unity beta directly in this case, we are getting the transmission scalar to equate with a homogeneous model *with the same transmission rate*, $c\beta_{\text{het}}$. But we are actually interested in finding the transmission scalar to equate with a homogeneous model *with the same number of infections at equilibrium*, β_{hom}. Thus, we have to adjust unity beta to account for the difference in trnasmission rate and the difference in the total number of susceptible individuals at equilibrium (also note, we find the scalar numerically, so there will be slight differences in number of infections as well).

Thus, we have

$$\text{age-structured infections} = \frac{\beta c}{N} \sum_{i=1}^{n} \sum_{i=j}^{n} \theta_{i,j}^{\text{het}} S_i^{\text{het}} I_j^{\text{het}} \tag{12}$$

$$\text{homogenous infections} = \frac{\beta}{N} \sum_{i=1}^{n} \sum_{i=j}^{n} \theta_{i,j}^{\text{hom}} S_i^{\text{hom}} I_j^{\text{hom}} \tag{13}$$

WLOG, let $\theta_{i,j}^{\text{hom}} = 1$ for all i, j. We can then equate these two quantities with unity beta scalar k:

$$\frac{\beta}{N} \sum_{i=1}^{n} \sum_{i=j}^{n} S_i^{\text{hom}} I_j^{\text{hom}} = k \frac{\beta c}{N} \sum_{i=1}^{n} \sum_{i=j}^{n} \theta_{i,j}^{\text{het}} S_i^{\text{het}} I_j^{\text{het}} \tag{14}$$

where

$$k = \frac{\sum_{i=1}^{n} S_i^{\text{hom}} \sum_{i=j}^{n} I_j^{\text{hom}}}{c \sum_{i=1}^{n} \sum_{i=j}^{n} \theta_{i,j}^{\text{het}} S_i^{\text{het}} I_j^{\text{het}}} \tag{15}$$

This is a scaled ratio of the number of new infections in the homogenous model to the number of new infections in the age-structured model. Interpretation is the same as the unity beta above, and showing on the log scale allows for equal visual comparison between n-fold reductions and increases in transmssion rate.

2.2 R_0 of the age-structured: using Next Generation Matrix method

To calculate R_0 in the age-structured model, we use the Next Generation Matrix (NGM) method outlined in Diekmann et al 1990. To do so, we follow the steps from the Bjornstad book:

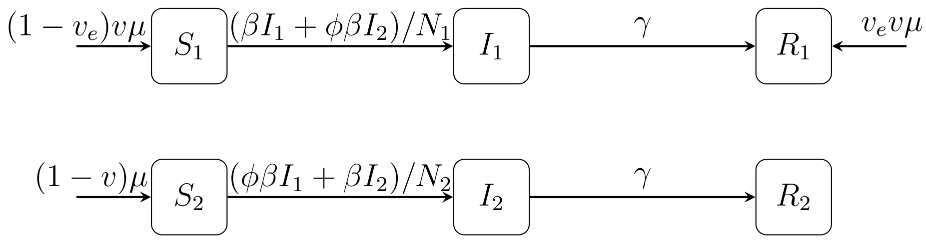
1. Define the infected states, $X = (I_1, I_2, ..., I_n)$ for ages $i = 1, 2, ..., n$
2. Define the new infection terms, $\mathcal{F}(X) = \left[\frac{\beta}{N}\sum_{i=j}^{n}\theta_{i,j}I_jS_i\right]$ for ages $i = 1, 2, ..., n$
3. Define all losses out of infected states, $\mathcal{V}^{-}(X) = \left[(\gamma+\mu)I_i\right]$ repeated for all ages.
4. Define all gained transfers between infected states, $\mathcal{V}^{+}(X) = \left[0\right]$ repeated for all ages.
5. Define the total losses out of infected states, $\mathcal{V}(X) = \mathcal{V}^{-}(X) - \mathcal{V}^{+}(X) = \left[(\gamma+\mu)\right]$.
6. Compute the jacobians of F and V. We do not evaluate at the disease-free equilibrium, because we will also use this evaluated at to calculate R_t. The jacobian of V, $v = \left[\frac{\partial \mathcal{F}_i}{\partial X_j}\right]$, will be $v = (\gamma+\mu)$ along the diagonal and zero elsewhere. The jacobian of F is

$$f = \left[\frac{\partial \mathcal{F}_i}{\partial X_j}\right] = \frac{\beta}{N}\begin{bmatrix} \theta_{1,1}S_1 & \theta_{1,2}S_1 & \cdots & \theta_{1,n}S_1 \\ \theta_{2,1}S_2 & \theta_{2,2}S_2 & \cdots & \theta_{2,n}S_2 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{n,1}S_n & \theta_{n,2}S_n & \cdots & \theta_{n,n}S_n \end{bmatrix}$$

7. Compute the next generation matrix, $K = fv^{-1}$. Because v is diagnoal, then v^{-1} is another diagonal matrices with elements $1/(\gamma+\mu)$. Then, R_0 is the largest eigenvalue of K, evaluated at the disease-free equilibrium (or with $S(t)$ for R_t).

3 Two-patch model

Next, we consider how the honeymoon period is affected by heterogeneity in the population, between vaccinated and unvaccinated individuals. We will use a two-patch model to illustrate this. Consider two subpopulations, representing the vaccinated and unvaccinated groups. Within each subpopulation, we have age-structured SIR dynamics, and ϕ governs the amount of mixing between the two groups. We also consider an imperfect vaccine, where v_e is the vaccine efficacy. (Age structure is not shown in the diagram for simplicity).



We can write the models equations as follows:

$$\frac{dS_1}{dt} = (1-v_1)\mu N_1 - (\beta I_1 + \phi\beta I_2)S_1/N_1 - \mu S_1 \tag{16}$$
$$\frac{dI_1}{dt} = (\beta I_1 + \phi\beta I_2)S_1/N_1 - (\gamma+\mu)I_1 \tag{17}$$
$$\frac{dR_1}{dt} = v_1\mu N_1 + \gamma I_1 - \mu R_1 \tag{18}$$
$$\frac{dS_2}{dt} = (1-v_2)\mu N_2 - (\phi\beta I_1 + \beta I_2)S_2/N_2 - \mu S_2 \tag{19}$$
$$\frac{dI_2}{dt} = (\phi\beta I_1 + \beta I_2)S_2/N_2 - (\gamma+\mu)I_2 \tag{20}$$
$$\frac{dR_2}{dt} = v_2\mu N_2 + \gamma I_2 - \mu R_2 \tag{21}$$

3.1 R_0 of the two patch model: using Next Generation Matrix method

To calculate R_0 in the age-structured, two patch model, we again use the Next Generation Matrix (NGM) method outlined in Diekmann et al 1990. To do so, we follow the steps from the Bjornstad book:

1. Define the infected states, $X = (I_{1,1}, I_{1,2}, \ldots, I_{1,n}, I_{2,1}, I_{2,2}, \ldots, I_{2,n})$ for ages $i = 1, 2, \ldots n$

2. Define the new infection terms, for ages $i = 1, 2, \ldots, n$

$$\mathcal{F}(X) = \begin{bmatrix} (I_{1,1} + \phi I_{2,1})\frac{S_{1,1}}{N_1} \\ (I_{1,2} + \phi I_{2,2})\frac{S_{1,2}}{N_1} \\ \vdots \\ (I_{1,n} + \phi I_{2,n})\frac{S_{1,n}}{N_1} \\ (\phi I_{1,1} + I_{2,1})\frac{S_{2,1}}{N_2} \\ (\phi I_{1,2} + I_{2,2})\frac{S_{2,2}}{N_2} \\ \vdots \\ (\phi I_{1,n} + I_{2,n})\frac{S_{2,n}}{N_2} \end{bmatrix}$$

3. Define all losses out of infected states, $\mathcal{V}^-(X) = \Big[(\gamma+\mu)I_{1,i}, (\gamma+\mu)I_{2,i}\Big]^T$ repeated for all ages.

4. Define all gained transfers between infected states, $\mathcal{V}^+(X) = \Big[0\Big]$ repeated for all ages.

5. Define the total losses out of infected states, $\mathcal{V}(X) = \mathcal{V}^-(X) - \mathcal{V}^+(X) = \Big[(\gamma+\mu)I\Big]$.

6. Compute the jacobians of F and V. We do not evaluate at the disease-free equilibrium, because we will also use this evaluated at to calculate R_t. The jacobian of V, $v = \left[\frac{\partial \mathcal{F}_i}{\partial X_j}\right]$, will be $v = (\gamma+\mu)$ along the diagonal and zero elsewhere. The jacobian of F is

$$f = \left[\frac{\partial \mathcal{F}_i}{\partial X_j}\right] = \beta \begin{bmatrix} \theta_{1,1}\frac{S_{1,1}}{N_1} & \cdots & \theta_{1,n}\frac{S_{1,1}}{N_1} & \phi\theta_{1,1}\frac{S_{1,1}}{N_1} & \cdots & \phi\theta_{1,n}\frac{S_{1,1}}{N_1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \theta_{n,1}\frac{S_{1,n}}{N_1} & \cdots & \theta_{n,n}\frac{S_{1,n}}{N_1} & \phi\theta_{n,1}\frac{S_{1,n}}{N_1} & \cdots & \phi\theta_{n,n}\frac{S_{1,n}}{N_1} \\ \phi\theta_{1,1}\frac{S_{2,1}}{N_2} & \cdots & \phi\theta_{1,n}\frac{S_{2,1}}{N_2} & \theta_{1,1}\frac{S_{2,1}}{N_2} & \cdots & \theta_{1,n}\frac{S_{2,1}}{N_2} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \phi\theta_{n,1}\frac{S_{2,n}}{N_2} & \cdots & \phi\theta_{n,n}\frac{S_{2,n}}{N_2} & \theta_{n,1}\frac{S_{2,n}}{N_2} & \cdots & \theta_{n,n}\frac{S_{2,n}}{N_2} \end{bmatrix}$$

7. Compute the next generation matrix, $K = fv^{-1}$. Because v is diagnoal, then v^{-1} is another diagonal matrices with elements $1/(\gamma+\mu)$. Then, R_0 is the largest eigenvalue of K, evaluated at the disease-free equilibrium (or with $S(t)$ for R_t).

IN this case, the next generation matrix can be written in block form as

$$N = \left[\frac{\partial \mathcal{F}_i}{\partial X_j}\right] = \frac{\beta}{\gamma+\mu}\begin{bmatrix} A & \phi A \\ \phi B & B \end{bmatrix}$$

Now, we will consider the simplest case where the age structure and epidemiological parameters are identical in both patches. However, we will assume that vaccination coverage is different in each patch. Thus, our next generation matrix can be written as

$$N = \frac{\beta}{\gamma+\mu}\begin{bmatrix} (1-v_1)C & \phi(1-v_1)C \\ (1-v_2)\phi C & (1-v_2)C \end{bmatrix} = \begin{bmatrix} (1-v_1) & \phi(1-v_1) \\ (1-v_2)\phi & (1-v_2) \end{bmatrix} * C$$

where $C = \theta diag(A)$ is the product of the WAIFW matrix θ and a diagonal matrix with entries of the stable age distribution, $diag(A)$. Then, let λ_C be an eigenvalue for C, such that $\lambda_C x = Cx$. Then, we can consider an eigenvector of the form $v = \begin{bmatrix} x \\ \alpha x \end{bmatrix}$.
Ultimately, we want to find the dominant eigenvalue λ of the next generation matrix, such that

$$Nx = \frac{\beta}{\gamma+\mu}\begin{bmatrix} (1-v_1) & \phi(1-v_1) \\ (1-v_2)\phi & (1-v_2) \end{bmatrix} * C * x = \lambda x$$

We will considering an eigenvector of the form $v = \begin{bmatrix} x \\ \alpha x \end{bmatrix}$, which implies that

$$\frac{\beta}{\gamma+\mu}\begin{bmatrix} (1-v_1) & \phi(1-v_1) \\ (1-v_2)\phi & (1-v_2) \end{bmatrix} \lambda_C \begin{bmatrix} x \\ \alpha x \end{bmatrix} = \lambda \begin{bmatrix} x \\ \alpha x \end{bmatrix}$$

which gives (ignoring $\beta/(\gamma+\mu)$ for now)

$$(1-v_1)\lambda_C x + \phi(1-v_1)\lambda_C \alpha x = \lambda x \tag{22}$$

$$(1-v_2)\phi\lambda_C x + (1-v_2)\lambda_C \alpha x = \lambda \alpha x \tag{23}$$

Assuming that $x \neq [0]$, we have

$$(1-v_1)\lambda_C + \phi(1-v_1)\lambda_C \alpha = \lambda \tag{24}$$

$$(1-v_2)\phi\lambda_C + (1-v_2)\lambda_C \alpha = \lambda \alpha \tag{25}$$

and this implies

$$\alpha = \frac{\lambda - (1-v_1)\lambda_C}{\phi(1-v_1)\lambda_C} \tag{26}$$

$$\alpha \frac{(1-v_2)\phi\lambda_C}{\lambda - (1-v_2)\lambda_C} \tag{27}$$

or

$$\frac{(1-v_2)\phi\lambda_C}{\lambda - (1-v_2)\lambda_C} = \frac{\lambda - (1-v_1)\lambda_C}{\phi(1-v_1)\lambda_C} \tag{28}$$

$$(1-v_1)(1-v_2)\phi^2\lambda_C^2 = \lambda^2 - \lambda(1-v_1)\lambda_C - \lambda(1-v_2)\lambda_C + (1-v_1)(1-v_2)\lambda_C^2 \tag{29}$$

$$0 = \lambda^2 - [(1-v_1)\lambda_C - (1-v_2)\lambda_C]\lambda + (1-v_1)(1-v_2)\lambda_C^2(1-\phi^2) \tag{30}$$

$$\tag{31}$$